

EXERCISE-1

Q1. Installing node latest version and npm

-Installing npm:

Step 1: In terminal:

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash
```

Step 2: Verify installation:

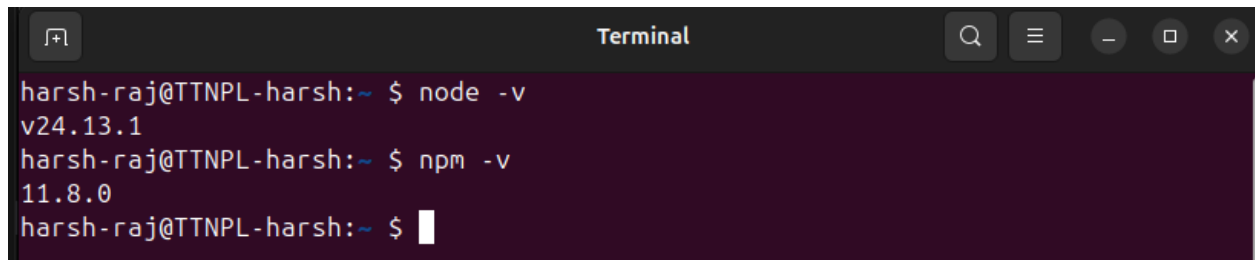
```
nvm --version
```

-Installing node:

Step 1: In terminal:

```
nvm install node
```

Step 2: node -v

A terminal window titled "Terminal" with a dark background. It shows the output of running 'node -v' and 'npm -v' in a shell. The prompt is 'harsh-raj@TTNPL-harsh:~ \$'.

```
harsh-raj@TTNPL-harsh:~ $ node -v
v24.13.1
harsh-raj@TTNPL-harsh:~ $ npm -v
11.8.0
harsh-raj@TTNPL-harsh:~ $
```

Q2. Read file asynchronously and synchronously

-Asynchronous:

Code:

A code editor showing a JavaScript snippet for asynchronous file reading. The code is numbered 1 to 16.

```
1  const fs=require("fs");
2
3  const read=>(){
4      console.log('Start');
5      fs.readFile('./file.txt', 'utf-8',function(err,content){
6          if(err){
7              console.log(err);
8              return;
9          }
10         console.log(content);
11     });
12
13     console.log('end');
14 }
15
16 read();
```

Output:

```
harsh-raj@TTNPL-harsh:Exercise1 $ node q1_async.js
Start
end
Hello this is me Harsh Raj
harsh-raj@TTNPL-harsh:Exercise1 $
```

-Synchronous:

Code:

```
Exercise1 >  q1_syncjs > ...
1  const fs= require("fs")
2
3  const readSync()=>{
4      console.log('Start')
5
6      const content=fs.readFileSync('./file.txt','utf-8');
7      console.log(content);
8      console.log('end');
9  }
10
11  readSync();
```

Output:

```
harsh-raj@TTNPL-harsh:Exercise1 $ node q1_sync.js
Start
Hello this is me Harsh Raj
end
```

Q3. Install dependencies and devdependencies.

Dependencies Installation:

```
● harsh-raj@TTNPL-harsh:Exercisel $ npm install express  
  
added 65 packages, and audited 66 packages in 6s  
  
22 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities  
○ harsh-raj@TTNPL-harsh:Exercisel $
```

DevDependencies Installation:

```
● harsh-raj@TTNPL-harsh:Exercisel $ npm install nodemon --save-dev  
  
added 26 packages, and audited 92 packages in 2s  
  
27 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities
```

Q4. Creating module and import(require) & export

Code:

stringUtils

```
Exercise1 > modules > JS stringUtils.js > [?] <unknown>
1  function toUpper(text){
2      return text.toUpperCase()
3  }
4
5  function countCharacters(text){
6      return text.length;
7  }
8
9  module.exports={toUpper,countCharacters}
```

Import stringUtils in app.js file

```
Exercise1 > JS app.js > ...
1  const stringUtils =require("./modules/stringUtils")
2
3  console.log(stringUtils.toUpper("node js assignment"));
4  console.log("length:" + stringUtils.countCharacters("node js"));
```

Output:

```
● harsh-raj@TTNPL-harsh:Exercise1 $ node app.js
NODE JS ASSIGNMENT
length:7
```

Q5. Creating an http server

Code:

```
Exercise1 > .js server.js > ...  
1  const http=require('http');  
2  
3  const server=http.createServer((req,res)=>{  
4    res.end("Hello this is me, Harsh")  
5  })  
6  server.listen(3000);
```

Server:

